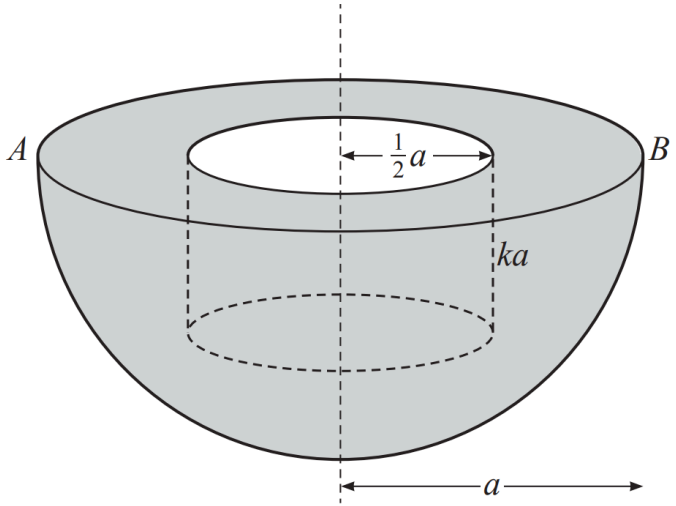
(FM) 2. Equilibrium of a rigid body HW

NO. 1



An object is formed by removing a solid cylinder, of height ka and radius $\frac{1}{2}a$, from a uniform solid hemisphere of radius a. The axes of symmetry of the hemisphere and the cylinder coincide and one circular face of the cylinder coincides with the plane face of the hemisphere. AB is a diameter of the circular face of the hemisphere (see diagram).

a. Show that the distance of the centre of mass of the object from AB is $\frac{3a(2-k^2)}{2(8-3k)}$. [4]

When the object is freely suspended from the point A, the line AB makes an angle θ with the downward vertical, where $\tan\theta = \frac{7}{18}$.

b. Find the possible values of k. [3]

A level FURTHER MATHEMATICS 2021/Winter/Paper3_2/Q4

Mark scheme

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td rowspan="4">a</td><td>
<table>
<tr><td></td><td>Volume</td><td>Centre of mass from AB</td></tr>
<tr><td>Hemisphere</td><td>$\frac{2}{3}\pi a^3$</td><td>$\frac{3}{8}a$</td></tr>
<tr><td>Cylinder</td><td>$\pi ka(\frac{a}{2})^2$</td><td>$\frac{ka}{2}$</td></tr>
<tr><td>Remainder</td><td>$\frac{2}{3}\pi a^3 - \pi ka\left(\frac{a}{2}\right)^2$</td><td>$\overline{x}$</td></tr>
</table>
</td><td>M1</td><td>Attempt at moments, 3 terms.</td></tr>
<tr><td>Taking moments about AB:
$\left(\frac{2}{3}\pi a^3 - \pi ka\left(\frac{a}{2}\right)^2\right) \times \overline{x} = \left(\frac{2}{3}\pi a^3 \times \frac{3}{8}a\right) - \left(\pi ka\left(\frac{a}{2}\right)^2 \times \frac{ka}{2}\right)$</td><td>A1
A1</td><td>Any 2 terms correct.
All correct.</td></tr>
<tr><td>$\overline{x} = \frac{3a\left(2-k^2\right)}{2(8-3k)}$</td><td>A1</td><td>Shown convincingly, AG.</td></tr>
<tr><td></td><td>4</td><td></td></tr>
<tr><td rowspan="4">b</td><td>$\tan\theta = \frac{\overline{x}}{a}$
$\frac{3\left(2-k^2\right)}{2(8-3k)} = \frac{7}{18}$</td><td>B1</td><td></td></tr>
<tr><td>$27k^2 - 21k + 2 = 0$</td><td>M1</td><td>Rearrange to form quadratic.</td></tr>
<tr><td>$k = \frac{2}{3}$ and $k = \frac{1}{9}$</td><td>A1</td><td>Both answers correct.</td></tr>
<tr><td></td><td>3</td><td></td></tr>
</table>

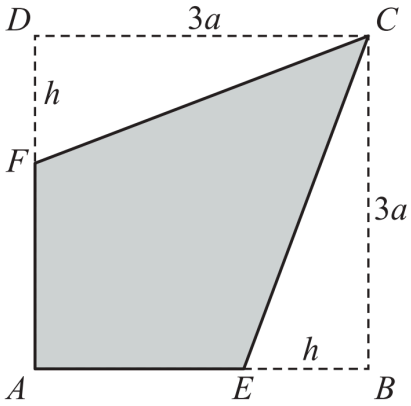
Comment

This question was answered well by the majority of candidates, with many excellent, clear and concise solutions.

a. Candidates had few difficulties in applying the method for finding the centre of mass of a solid made up of two parts to this situation. An error that occurred sometimes was to use the volume of a sphere instead of that of a hemisphere. Some candidates did not convincingly show the given result. In these cases there was an unreasonable jump from the penultimate line of working to the given result. Candidates are reminded that when an answer is given in the question, they should give sufficient working to convince the Examiner that they have achieved the given result.

b. Most candidates understood the geometry of the situation, wrote down the expression $\tan\theta = \frac{\bar{x}}{a}$ and used the result of **part (a)** to find the two correct values of k. A few candidates had the expression for $\tan\theta$ inverted, and others made algebraic slips when simplifying.

NO. 2



A uniform lamina $AECF$ is formed by removing two identical triangles BCE and CDF from a square lamina $ABCD$. The square has side $3a$ and $EB = DF = h$ (see diagram).

a. Find the distance of the centre of mass of the lamina $AECF$ from AD and from AB, giving your answers in terms of a and h. [5]

The lamina $AECF$ is placed vertically on its edge AE on a horizontal plane.

b. Find, in terms of a, the set of values of h for which the lamina remains in equilibrium. [3]

A level FURTHER MATHEMATICS 2021/Winter/Paper3_1/Q4

Mark scheme

Question	Answer	Marks	Guidance
a	<table><tr><th></th><th>Area</th><th>Centre of mass from AD</th></tr><tr><td>Square</td><td>$9a^2$</td><td>$\frac{3}{2}a$</td></tr><tr><td>CDF</td><td>$\frac{3}{2}ah$</td><td>a</td></tr><tr><td>BEC</td><td>$\frac{3}{2}ah$</td><td>$3a-\frac{1}{3}h$</td></tr><tr><td>Resulting $AEFC$</td><td>$9a^2-3ah$</td><td></td></tr></table>	M1	Attempt at moments with three terms.
	Taking moments about AD: $\left(9a^2-3ah\right)\ \bar{x}=\left(9a^2\times\frac{3}{2}a\right)-\left(\frac{3}{2}ah\times a\right)-\left(\frac{3}{2}ah\times\left(3a-\frac{1}{3}h\right)\right)$	A1 A1	Two terms correct. All correct.
	$\bar{x}=\frac{27a^2-12ah+h^2}{6(3a-h)}\ \left(=\frac{9a-h}{6}\right)$	A1	AEF
	$\bar{y}=\bar{x}$	B1	By symmetry or equal to their $\bar{x}$.
		5	
b	For equilibrium, $\bar{x}\leqslant 3a-h$ $27a^2-12ah+h^2\leqslant 6(3a-h)^2$	B1	Accept strict inequality.
	$27a^2-24ah+5h^2\geqslant 0$	M1	Homogeneous 3-term quadratic inequality.
	$h\leqslant\frac{9}{5}a$	A1	CAO.

		3	

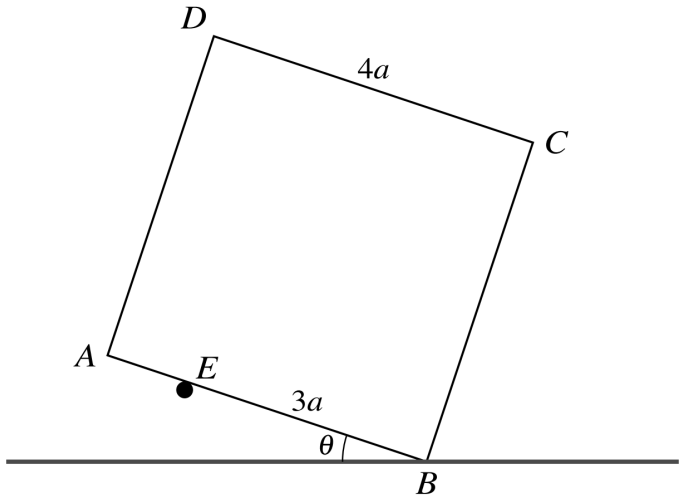
Comment

a. This part question was answered correctly by many candidates. They correctly calculated the moments and confidently manipulated the equation to obtain the correct answer. A typical error was to express the distance of the centre of mass of triangle *BEC* from the line segment *AD* as $\left(3a-\frac{2}{3}h\right)$ instead of $\left(3a-\frac{1}{3}h\right)$. A few candidates factorised the numerator of their answer and simplified it to obtain $\bar{x}=\frac{9a-h}{6}$, which greatly simplified their work in **part (b)**.

Some candidates obtained the last mark in an elegant and efficient way by stating that $\bar{x}=\bar{y}$ because of the symmetry of the lamina. Other candidates did not spot this symmetry and calculated $\bar{y}$ using the same approach as for $\bar{x}$, sometimes incorrectly.

b. Some candidates annotated the given diagram to answer this part question. Most of them identified the correct inequality $\bar{x}\leqslant 3a - h$ and then obtained the correct answer. The most common mistake in this case was to use a strict inequality.

NO. 3



A uniform square lamina $ABCD$ of side $4a$ and weight W rests in a vertical plane with the edge AB inclined at an angle θ to the horizontal, where $\tan\theta = \frac{1}{3}$. The vertex B is in contact with a rough horizontal surface for which the coefficient of friction is μ. The lamina is supported by a smooth peg at the point E on AB, where $BE = 3a$ (see diagram).

i. Find expressions in terms of W for the normal reaction forces at E and B. [5]

ii. Given that the lamina is about to slip, find the value of μ. [3]

A level FURTHER MATHEMATICS 2019/Winter/Paper2_1/Q2

Mark scheme

Question	Answer	Marks	Guidance
2.i	$R_E \times 3a = W\cos\theta \times 2a - W\sin\theta \times 2a$ or $R_E \times 3a = W \times 2a\,(1 - \tan\theta)\cos\theta$ or $R_E \times 3a = 2\sqrt{2}W\sin\left(\frac{\pi}{4} - \theta\right)$	**M1 A1**	Take moments about B
	$R_E = 4W / 3\sqrt{10}$	**A1**	Find normal reaction at E. AEF
	$R_B = W - R_E\cos\theta = 3W/5$	**M1 A1**	Find normal reaction at B by resolving forces vertically
		5	
ii	$F_B = R_E\sin\theta = 2W/15$	**M1 A1**	Find friction at B by resolving forces horizontally
	$\mu = (2/15) / (3/5) = 2/9$	**A1**	Find μ from $F_B = \mu R_B$
		3	

Comment

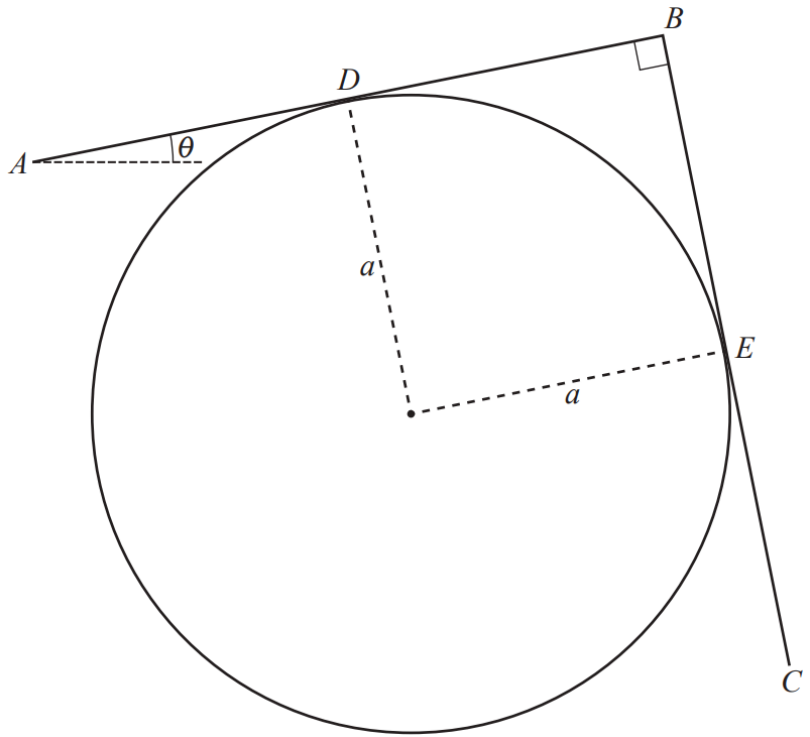
As in all questions of this type, candidates are well advised to first identify all the forces acting on the lamina, preferably showing them on the given diagram. This will help the candidate to ensure that they include all the relevant forces when taking moments or resolving. Candidates were required to find the normal reaction forces at E and B. No method was suggested, so the candidate was free to choose the approach. The most direct method of solution is to take moments about the point B, thereby eliminating all the forces except the reaction at E and the weight acting at the centre of the lamina. The second of these proved troublesome to many candidates, who had difficulty in finding the distance of the line of action of the weight from the point B.

The simplest way was to resolve the weight into two components, one parallel to AD and one perpendicular to AD. Alternatively, a distance involving a sine or cosine of a sum of two angles, and its expansion, is required. Using this approach, the next step is to resolve the forces vertically, leading to an expression for the normal reaction force at E, and then horizontally to find the frictional force.

Of course it is possible to take moments about several other points, and many candidates did indeed do so. Almost invariably, however, the candidates who did this were unsuccessful in isolating the forces that they required from the resulting moments

equations.

NO. 4



A uniform cylinder with a rough surface and of radius a is fixed with its axis horizontal. Two identical uniform rods AB and BC, each of weight W and length $2a$, are rigidly joined at B with AB perpendicular to BC. The rods rest on the cylinder in a vertical plane perpendicular to the axis of the cylinder with AB at an angle θ to the horizontal. D and E are the midpoints of AB and BC respectively and also the points of contact of the rods with the cylinder (see diagram). The rods are about to slip in a clockwise direction. The coefficient of friction between each rod and the cylinder is μ.

The normal reaction between AB and the cylinder is R and the normal reaction between BC and the cylinder is N.

a. Find the ratio $R : N$ in terms of μ. [6]

b. Given that $\mu = \frac{1}{3}$, find the value of $\tan\theta$. [3]

A level FURTHER MATHEMATICS 2022/Summer/Paper3_3/Q7

Mark scheme

Question	Answer	Marks	Guidance
a	Frictional force = μ × normal reaction at D and E	B1	$F_{AB} = \mu R$, $F_{BC} = \mu N$
	Moments about B, $Na - Ra = Wa(\sin\theta - \cos\theta)$ Moments about centre, $F_{AB}a + F_{BC}a = Wa(\cos\theta - \sin\theta)$ Moments about D, $F_{BC}a + Na = Wa(\cos\theta + \sin\theta)$ Moments about E, $Ra - F_{AB}a = Wa(\cos\theta + \sin\theta)$	B1	One moments equation about any point involving all relevant forces, resolved if necessary (AEF).
	Parallel to AB, $Na - F_{AB} = W\sin\theta + W\sin\theta$ Perpendicular to AB, $F_{BC} + R = W\cos\theta + W\cos\theta$	B1	Two resolutions: all relevant terms, different frictional forces [Vertical: $R\cos\theta + F_{BC}\cos\theta + N\sin\theta = F_{AB}\sin\theta + W + W$ Horizontal: $F_{BC}\sin\theta + F_{AB}\cos\theta + R\sin\theta$] Alternative approach using two moments equations can earn the B1B1
	$N - R = \frac{1}{2}\left((1-\mu)N - (1+\mu)R\right)$	M1	Combine appropriate equations.
	$N\left(1 - \frac{1}{2}(1-\mu)\right) = R\left(1 - \frac{1}{2}(1+\mu)\right)$ $N\left(\frac{1}{2} + \frac{1}{2}\mu\right) = R\left(\frac{1}{2} - \frac{1}{2}\mu\right)$	M1	Collect terms to obtain ratio/fraction in terms of μ only (CWO), any equivalent simplified form.
	$R : N = 1 + \mu : 1 - \mu$	A1	
		6	
b	Divide resolution equations: $\tan\theta = \frac{N - \mu R}{\mu N + R}$	M1	Must include μ terms.

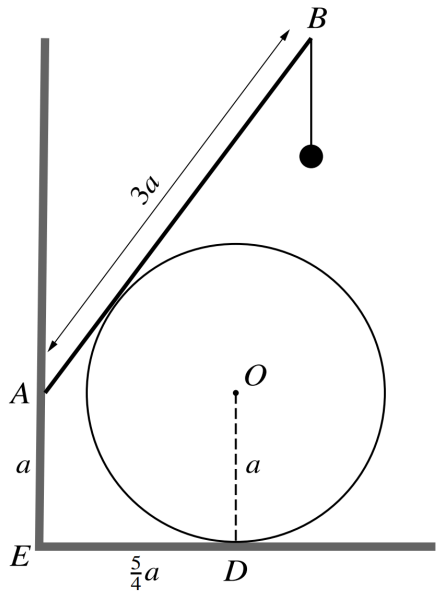
	Use $R = 2N$: $\tan\theta = \dfrac{\frac{1}{3}N}{\frac{7}{3}N}$	M1	FT their answer to part **(a)**.
	$\tan\theta = \dfrac{1}{7}$	A1	
		3	

Comment

a. The majority of candidates annotated the given diagram with all the relevant forces at D and E, in the relevant directions: normal reactions, frictional forces and weights. They then had to make choices about resolving forces and forming moments equations. The most efficient method is to take moments about D and about E. From these two equations, the relationship between N and R can be found. Of course, moments can be taken about any point, although the resulting equations will not all be quite as simple. Another approach is to take moments about one point and then also resolve in two directions for the whole system.
A very common error was to resolve perpendicular to each rod, individually, giving the equations $R = W\cos\theta$, $N = W\sin\theta$. These results are not correct as they are based on a misunderstanding that parts of the system can be considered in isolation. This is not so since all resolving of forces must include all the forces acting on the whole rigid body. The majority of candidates adopted this invalid approach and as a result very few fully correct solutions were seen.

b. Very few candidates answered this part successfully.

NO. 5



A uniform smooth disc with centre O and radius a is fixed at the point D on a horizontal surface. A uniform rod of length $3a$ and weight W rests on the disc with its end A in contact with a rough vertical wall. The rod and the disc lie in a vertical plane that is perpendicular to the wall. The wall meets the horizontal surface at the point E such that $AE = a$ and $ED = \frac{5}{4}a$. A particle of weight kW is hung from the rod at B (see diagram). The coefficient of friction between the rod and the wall is $\frac{1}{8}$ and the system is in limiting equilibrium. Find the value of k. [8]

A level FURTHER MATHEMATICS 2017/Summer/Paper2_1/Q2

Mark scheme

Question	Answer	Marks	Guidance
2	Take moments for rod about some point such as: A: $R_P \times A_P - kW \times 3a\cos\theta = W \times (3a/2)\cos\theta$ [$R_P \times 3a/4 - kW \times 9a/5 = W \times 9a/10$ so $15R_P - 36kW = 18W$] O: $F_A \times (5a/4) - kW \times (3a\cos\theta - 5a/4)$ $= -W \times (5a/4 - (3a/2)\cos\theta)$ [$F_A \times 5a/4 - kW \times 11a/20 = -W \times 7a/20$ so $25F_A - 11kW = -7W$] P: $R_A \times AP\sin\theta + F_A \times AP\cos\theta - kW \times (3a - AP)\cos\theta$ $= W \times (3a/2 - AP)\cos\theta$) [$R_A \times 3a/5 + F_A \times 9a/20 - kW \times 27a/20 = W \times 9a/20$ so $12R_A + 9F_A - 27kW = 9W$] B: $R_P \times (3a - AP) - R_A \times 3a\sin\theta - F_A \times 3a\cos\theta$ $= W \times (3a/2)\cos\theta$ [$R_P \times 9a/4 - R_A \times 12a/5 - F_A \times 9a/5 = W \times 9a/10$ so $45R_P - 48R_A - 36F_A = 18W$] C: $R_P \times (3a/2 - AP) - R_A \times (3a/2)\sin\theta - F_A \times (3a/2)\cos\theta + kW \times (3a/2)\cos\theta = 0$ [$R_P \times 3a/4 - R_A \times 6a/5 - F_A \times 9a/10 + kW \times 9a/10 = 0$ so $15R_P - 24R_A - 18F_A + 18kW = 0$] F: $R_P\cos\theta \times (3a - AP)\cos\theta - R_P\sin\theta \times AP\sin\theta - F_A \times 3a\cos\theta = W \times (3a/2)\cos\theta$ [$(3/5)R_P \times 27a/20 - (4/5)R_P \times 3a/5 - F_A \times 9a/5 = W \times 9a/10$ so $81R_P - 48R_P - 180F_A = 90W$]	**M1A1**	F_A here denotes friction on rod measured in downward dirn; P denotes point of contact of rod and disc; θ denotes angle between rod and horizontal. [$AP = 3a/4$, $\sin\theta = 4/5$, $\cos\theta = 3/5$, $\tan\theta = 4/3$, $3a - AP = 9a/4$, $3a/2 - AP = 3a/4$] See note below on solving question without introducing R_P (C denotes mid-point of AB) (F is vertically below B, on AO extended)
	Find two more indep. eqns, e.g. resolution of forces on rod: Horizontally: $R_A = R_P\sin\theta$ [$= 4R_P/5$] Vertically: $F_A + (k + 1)W = R_P\cos\theta$ [$= 3R_P/5$]	**B1B1**	A second moment eqn. may be used instead of a resolution

	Along AB: $R_A \cos\theta = F_A \sin\theta + (k + 1)W \sin\theta$ Normal to AB: $R_P = R_A \sin\theta + F_A \cos\theta + (k + 1)W \cos\theta$		Count as 2 eqns if used with moments about P (so R_P absent)
	$F_A = +R_A / 8$ *or* $-R_A / 8$ as appropriate	**B1**	Relate F_A and R_A (may be implied; and must be consistent with friction taken down or up in above eqns)
	[$\sin\theta = 4/5$, $\cos\theta = 3/5$, $\tan\theta = 4/3$]	**M1**	Eliminate θ from all reqd. independent eqns. for forces Find either value of k from reqd. independent eqns. for forces
	$F_A \downarrow$: [$R_P = 6W$, $F_A = 3W/5$, $R_A = 24W/5$], $k = 2$ $F_A \uparrow$: [$R_P = 30W/17$, $F_A = 3W/17$, $R_A = 24W/17$], $k = 4/17$	**M1A1**	(or 0.235)
	Total:	**8**	

Explanation

2.

Take moments for rod about some point such as:

A: $R_P \times A_P - kW \times 3a \cos\theta = W \times (3a/2) \cos\theta$

[$R_P \times 3a/4 - kW \times 9a/5 = W \times 9a/10$

so $15R_P - 36kW = 18W$]

O: $F_A \times (5a/4) - kW \times (3a \cos\theta - 5a/4)$

$= - W \times (5a/4 - (3a/2) \cos\theta)$

[$F_A \times 5a/4 - kW \times 11a/20 = - W \times 7a/20$

so $25F_A - 11kW = - 7W$]

P: $R_A \times AP \sin\theta + F_A \times AP \cos\theta - kW \times (3a - AP) \cos\theta$

$= W \times (3a/2 - AP) \cos\theta$)

[$R_A \times 3a/5 + F_A \times 9a/20 - kW \times 27a/20 = W \times 9a/20$

so $12R_A + 9F_A - 27kW = 9W$]

B: $R_P \times (3a - AP) - R_A \times 3a \sin\theta - F_A \times 3a \cos\theta$ M1 A1

$= W \times (3a/2) \cos\theta$

[$R_P \times 9a/4 - R_A \times 12a/5 - F_A \times 9a/5 = W \times 9a/10$

so $45R_P - 48R_A - 36F_A = 18W$]

C: $R_P \times (3a/2 - AP) - R_A \times (3a/2)\sin\theta - F_A \times (3a/2)\cos\theta + kW \times (3a/2)\cos\theta = 0$

[$R_P \times 3a/4 - R_A \times 6a/5 - F_A \times 9a/10 + kW \times 9a/10 = 0$

so $15R_P - 24R_A - 18F_A + 18kW = 0$]

F: $R_P \cos\theta \times (3a - AP)\cos\theta - R_P \sin\theta \times AP \sin\theta$

$- F_A \times 3a \cos\theta = W \times (3a/2)\cos\theta$

[$(3/5)R_P \times 27a/20 - (4/5)R_P \times 3a/5$

$- F_A \times 9a/5 = W \times 9a/10$

so $81R_P - 48R_P - 180F_A = 90W$]

Find two more indep. eqns, e.g. resolution of forces on rod:

Horizontally: $R_A = R_P \sin\theta$ [$= 4R_P/5$]

Vertically: $F_A + (k + 1)W = R_P \cos\theta$ [$= 3R_P/5$]

Along AB: $R_A \cos\theta = F_A \sin\theta + (k + 1)W \sin\theta$ B1 B1 A second moment eqn. may be used instead of a resolution

Normal to AB: $R_P = R_A \sin\theta + F_A \cos\theta + (k + 1)W \cos\theta$

$F_A = +R_A / 8$ or $-R_A / 8$ as appropriate B1 Relate F_A and R_A (may be implied; and must be consistent with friction taken down or up in above eqns)

[$\sin\theta = 4/5$, $\cos\theta = 3/5$, $\tan\theta = 4/3$] M1 Eliminate θ from all reqd. independent eqns. for forces Find either value of k from reqd. independent eqns. for forces

$F_A \downarrow$: [$R_P = 6W$, $F_A = 3W/5$, $R_A = 24W/5$], $k = 2$

$F_A \uparrow$: [$R_P = 30W/17$, $F_A = 3W/17$, $R_A = 24W/17$], $k = 4/17$ M1 A1 (or 0.235)

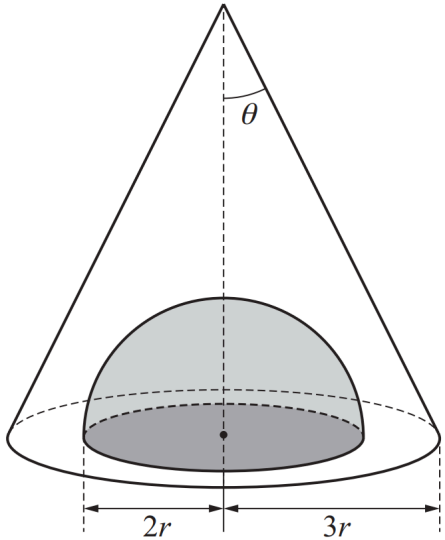
Comment

Candidates are well advised to first identify all the forces acting on the rod, preferably showing them on a diagram so that the symbols used to represent them are clear. This process reveals that three possible forces are unknown, namely the reaction R_A and friction F_A at A and the normal reaction R_P at P, the point of contact of the rod and disc. A further unknown is the required value of k, implying at first sight that four independent equations must be formulated in order to find k. One such follows immediately from the information about limiting equilibrium in the question, namely $F_A = R_A/8$. While there are a variety of possible moment and force resolution equations to choose from, many candidates chose to resolve forces vertically and horizontally and to take moments about A. It is convenient in this process to write the equations in terms of an angle such as OAP, and then replace in due course such factors as the sine and cosine of this angle, namely 4/5 and 3/5 respectively. Some candidates wrongly assumed from the diagram that the radius OP is parallel to DA, resulting in the incorrect values $5/\sqrt{41}$ and $4/\sqrt{41}$ being used for the above sine and cosine. As is sometimes the case in such questions, it is possible to reduce the work required by a careful choice of equations. Thus taking

moments about P, for example, together with only a resolution of forces along AB is sufficient since the unknown force R_P is not introduced. Finally, virtually all candidates gave no indication that the friction at A may act either upwards or downwards, giving different values for k, but full marks could be earned by a satisfactory consideration of only one case.

Answer: 4/17 or 2.

NO. 6



An object is formed by removing a solid hemisphere, radius $2r$, from a uniform solid cone, radius $3r$ and semi-vertical angle θ, where $\tan\theta = \frac{1}{2}$. The axes of symmetry of the cone and the hemisphere coincide. The base of the cone and the base of the hemisphere are in the same plane as each other (see diagram).

a. Find, in terms of r, the distance of the centre of mass of the object from its base. [4]

The object is placed such that its circular base makes contact with a rough plane which is inclined to the horizontal at an angle α. The object is on the point of toppling. The plane is sufficiently rough to prevent sliding.

b. Find the value of α. [3]

Mark scheme

Question	Answer	Marks	Guidance
a	<table><tr><th></th><th>Volume</th><th>CoM above base</th></tr><tr><td>Cone</td><td>$18\pi r^3$</td><td>$\frac{3}{2}r$</td></tr><tr><td>Hemisphere</td><td>$\frac{16}{3}\pi r^3$</td><td>$\frac{3}{4}r$</td></tr><tr><td>Total object</td><td>$18\pi r^3 - \frac{16}{3}\pi r^3$</td><td>$\bar{y}$</td></tr></table>	M1	Allow one error for cone or hemisphere. SOI, may be seen in moments equation.
	$\left(18\pi r^3 - \frac{16}{3}\pi r^3\right)\bar{y} = 18\pi r^3\left(\frac{3}{2}r\right) - \frac{16}{3}\pi r^3\left(\frac{3}{4}r\right)$	M1	A moments equation with 3 terms, allow sign errors, must be dimensionally correct.
		A1	Fully correct equation, does not need to be simplified.
	$\bar{y} = \frac{69}{38}r$	A1	
		4	
b	When on point of toppling: $\tan\alpha = \frac{3r}{their\ \bar{y}}$ $\left[=\frac{38}{23}\right]$	M1	For attempt at tanα using $\frac{3r}{their\ \bar{y}}$ or $\frac{their\ \bar{y}}{3r}$
		A1 FT	FT *their* $\bar{y}$.
	α = 58.8o	A1	CAO
		3	

Comment

a. Many candidates were successful in writing an accurate and dimensionally correct moments equation. Weaker responses often did not subtract the moment for the hemisphere from the moment for the cone.

b. The majority of candidates were able to provide a correct expression for the tangent of the required angle. Some responses used the reciprocal of the required fraction for $\tan\theta$. Drawing a sketch of the toppling situation with the object on an inclined plane could help avoid such errors and help weaker candidates make further progress.